

LLM Foundations Homework Sheet

In-class reflection and homework for Week 1 / Module 0: What Is an LLM?

Name

Date

Required workload: Class teaches the core concepts directly. The short 3Blue1Brown video is reinforcement and can be replayed for review. The main required homework is Karpathy's full 3h31m video. You may watch Karpathy at 1.25x or 1.5x, but pause at each checkpoint and answer in your own words.

Class Reinforcement and Required Homework

Optional Class Replay

3Blue1Brown: Large Language Models explained briefly

<https://www.youtube.com/watch?v=LPZh9BOjkQs>

Purpose: review the visual version of concepts already taught in class: next-token prediction, scale, chatbots, and training.

Required Homework Video

Andrej Karpathy: Deep Dive into LLMs like ChatGPT

<https://www.youtube.com/watch?v=7xTGNNLPyMI>

Purpose: full foundation for data, tokens, training, inference, hallucinations, tool use, RLHF, and practical use.

Optional reference: Karpathy's MicroGPT article shows the same GPT ingredients in a tiny setting: documents, tokenizer, model, training loop, and inference loop. Use it only if you want a code-level preview after the class concepts.

<https://karpathy.github.io/2026/02/12/microgpt/>

Concept Warm-Up Before the Required Karpathy Video

In one sentence, what do you think ChatGPT is doing when it answers a question?

Tiny Glossary

Term	Meaning for this course
Token	A chunk of text the model processes. It can be a word, part of a word, punctuation, or code fragment.
Context window	The current working memory: the text the model can see while answering.
Pretraining	Learning broad patterns from massive text and code data.
SFT	Supervised fine-tuning: training from example conversations or desired outputs.
RLHF	Reinforcement learning from human feedback: using human preferences to shape model behavior.
Inference	Running the trained model to generate output, usually one token at a time.
Hallucination	A fluent answer that is unsupported, misleading, or false.
Tool	An external action the system lets the model request, such as search, calculation, or code execution.
Guardrail	A rule, approval step, or check that limits risky behavior.

Class Concept Reflection

After the class concept slides and optional 3Blue1Brown replay, write two ideas that became clearer.

What is one question you still have before watching Karpathy?

Karpathy Checkpoints

Timestamps are based on the current YouTube chapters for "Deep Dive into LLMs like ChatGPT." Use them as anchors, not as exact quiz boundaries.

00:01:00 Pretraining data (internet)

What kinds of data help train an LLM? Why does data quality matter?

00:07:47 Tokenization

What is a token? Give one example where tokenization might surprise a user.

00:14:27 Neural network input/output

What goes into the model, and what comes out?

00:20:11 Neural network internals

You do not need the math. What is the useful mental model for what the network has learned?

00:26:01 Inference

Why does the answer appear one piece at a time?

00:31:09 GPT-2: training and inference

What does a base model do? How is that different from a helpful chat assistant?

00:42:52 Llama 3.1 base model inference

What did you notice about interacting with a base model?

00:59:23 Pretraining to post-training

Why is pretraining not enough if we want a useful assistant?

01:01:06 Post-training data (conversations)

How do conversation examples change the behavior of a model?

01:20:32 Hallucinations, tool use, knowledge, working memory

Why can a model sound confident and still be wrong? How can tools help?

01:41:46 Knowledge of self

Why might a model be confused about what it is, what version it is, or what it can access?

01:46:56 Models need tokens to think

Why can writing intermediate reasoning or scratch work help a model perform better?

02:01:11 Tokenization revisited

Why can spelling, counting letters, or manipulating strings be surprisingly hard?

02:04:53 Jagged intelligence

What does it mean for a model to be brilliant at one task and weak at another?

02:07:28 SFT to reinforcement learning

What changes when the system moves beyond examples and starts using feedback or rewards?

02:14:42 Reinforcement learning

Explain reinforcement learning in one sentence using your own words.

02:48:26 RLHF

Why do human preferences matter when building assistant-like models?

03:21:46 Grand summary

Write three takeaways you want to remember before learning agents.

Bridge to Agents

Use this course definition:

Practical agent = LLM + goal + context + tools + loop + guardrails

Classify each example as **chatbot answer**, **tool call**, or **agent loop**.

Example	Your classification	Why?
ChatGPT explains the difference between pretraining and post-training.		
A model asks a calculator tool to compute $438 * 27$.		

A coding system reads files, proposes a plan, edits code, runs tests, and asks before deploying.		
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Verification Habit

Pick one important claim from the video or from a chatbot answer. How would you verify it?

Exit Ticket

In 4-6 sentences, explain why an agent needs guardrails.

Completion Checklist

- ☐ I reviewed the class concepts and replayed the short 3Blue1Brown video if I needed the visual version.
- ☐ I watched the full Karpathy video, using the checkpoint questions.
- ☐ I completed the glossary and checkpoint responses in my own words.
- ☐ I can explain the difference between chatbot answer, tool call, and agent loop.
- ☐ I wrote an exit ticket about guardrails.

Teacher Grading Guide

Criteria	Points
Completed video checkpoints with specific, non-copy-pasted responses	40
Accurate use of core vocabulary: token, context, pretraining, post-training, inference, hallucination	20

Clear explanation of hallucination and verification behavior	15
Correctly distinguishes chatbot answer, tool call, and agent loop	15
Exit ticket explains why guardrails matter for risky actions	10

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